Variation in abundance and resilience of butterflies in Borneo

Methods

Weekly surveys were conducted over two years at four different monitoring sites on rare butterfly species across Borneo. This includes Sabah, Sarawak, West Kalimantan and East Kalimantan. Researchers recorded the amount of mature butterflies seen on the same standard one hour walk.

Due to the life span of the mature butterflies being less than one week each time they counted it represented a new cohort and could be considered as independent. Population resilience was calculated using the population growth rate (PGR), which was calculated by the change in the log of abundance between successive weekly counts.

Statistical analysis

All analyses were done in R Studio Version 4.4.2

Abundance difference across the different sites

Abundance data is skewed, and therefore a generalised linear model (GLM) was used to test whether abundance really was different between the sites.

The model did show that there was overdispersion (dispersion = 7.42), but the size between sites was very large, and the qualitative results are not significantly affected by the outliers and produce reliable results.

Resilience differences across the different sites

To test id the density dependence was different across the different sites, a linear model was used as seen below:

PGR ~ Abundance x Site

PGR was seen as a constant variable as the values were roughly balanced around zero.

Carrying capacity estimation

Carrying capacity (K) was worked out as the abundance regarding PGR = 0 for each of the sites. Separate regressions of the PGR compared to abundance were fitted for each site of Borneo.

Results

Abundance difference between sites

Butterfly abundance did vary severely among the four monitored regions (Figure 1). Sarawak displayed much higher counts (up to 140 individuals per hour) compared to the others, which seemed to generally show abundances below 20.

The generalised linear model shows a significant effect of site on abundance (Table 1). Compared to East Kalimantan, Sarawak had a 1.705 increase in log abundance, which is about a 5.5 times increase in expected density. Additionally, West Kalimantan also had a higher abundance (0.419, $p < 2x10\text{-}16$), while Sabah did not really change significantly from East

Kalimantan (0.060, p = 0.172). Even know dispersion = 7.42 and therefore was overly dispersed, the effect size was large enough that the results of strong special difference are fair.

Resilience difference between sites

The PGR showed the density across all the sites, and it can be seen that there are high growth rates at low abundance and negative growth rates at high abundances (Figure 2). The relation between abundance and site was strongly significant (46.04 on 7 and 404 DF, p-value: < 2.2e-16), showing that resilience differed among the four different populations.

Sarawak showed the most vertical negative slope (0.104, p < 2x10-16), reflecting strong density dependence. Sabah and East Kalimantan also showed negative relationships between abundance and the PGR, while on the other hand, West Kalimantan showed a weaker pattern (0.032, p = 0.008).

Carrying capacity

Carrying capacities from the site-specific regression showed large variation across the whole of Borneo (Table 2). West Kalimantan showed the highest estimated carrying capacity at 14.62, next was Sabah at 10.19, and then East Kalimantan at 9.61, while Sarawak had a very low estimate of only 1.81 due to a shallow regression slope. This shows the high variability in PGR at large population sizes.

Figures

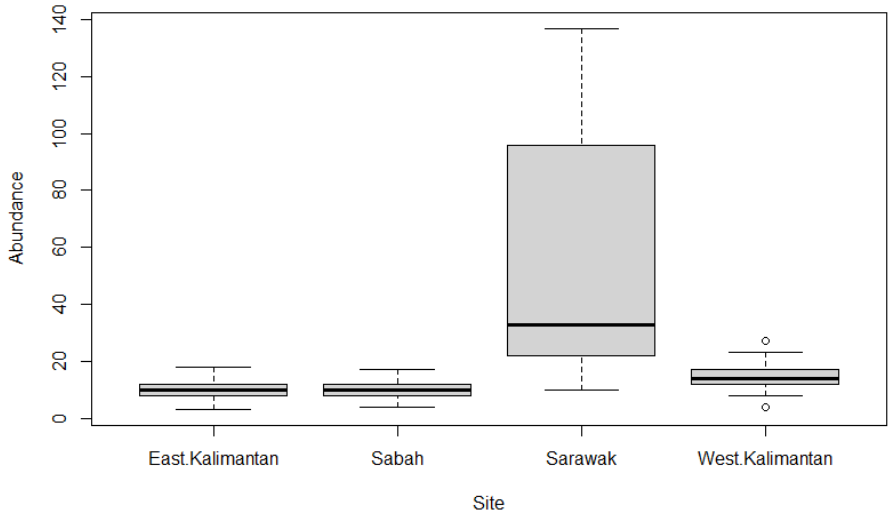


Figure 1: Abundance of butterfly species in the four sites monitored. Boxes show the medians and interquartile ranges, and the whiskers show 1.5x IQR

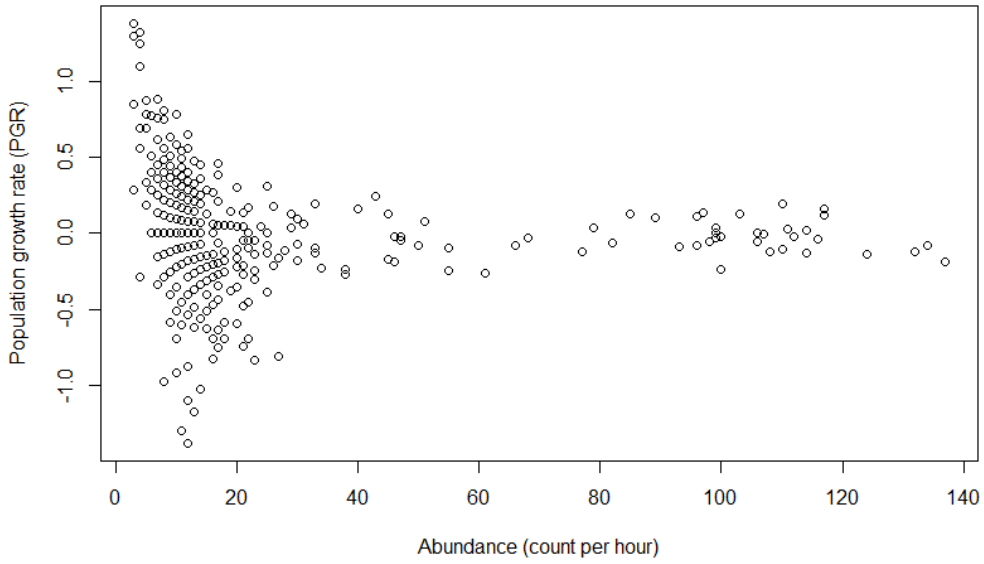


Figure 2: scatterplot of the population growth rate (PGR) against the abundance in all the sites

Tables

Table 1: Summary of the GLM testing for differences in abundance in the sites

Term	Estimate	standard error	Z value	P value
intercept	2.261	0.0316	71.44	$<2\times10^{-16}$
Sabah	0.06	0.0441	1.366	0.172
Sarawak	1.705	0.0344	49.56	$<2\times10^{-16}$
West Kalimantan	0.419	0.0408	10.29	$<2\times10^{-16}$

Table 2: estimated carrying capacities for each of the four sites

site	estimated K
Sabah	10.19
Sarawak	1.81
West Kalimantan	14.62
East Kalimantan	9.61

Appendix

```
butterfly_counts   <- j.c.bull_14112025_butterfly_counts

butterfly_dynamics <- j.c.bull_14112025_butterfly_dynamics

#str

str(butterfly_counts)
```

```
str(butterfly_dynamics)
# abundance differs between sites hyp1
# single response vector of counts
Abundance <- c(butterfly_counts$Sabah,
butterfly_counts$Sarawak,
butterfly_counts$West.Kalimantan,
butterfly_counts$East.Kalimantan)
#match site factor
Site <- c(rep("Sabah",length(butterfly_counts$Sabah)),
rep("Sarawak",length(butterfly_counts$Sarawak)),
rep("West.Kalimantan",length(butterfly_counts$West.Kalimantan)),
rep("East.Kalimantan",length(butterfly_counts$East.Kalimantan)))
Site <- factor(Site)
# plot da ting
plot(Abundance ~ Site)
#GLM for count data
m_abund <- glm(formula = Abundance ~ Site,
        family = poisson(link = "log"))
summary(m_abund)
#dispersion check for gang
residual_deviance <- summary(m_abund)$deviance
residual_df <- summary(m_abund)$df.residual
dispersion_abund <- residual_deviance / residual_df
dispersion_abund
# resilience differs between sites hyp2
#Combine abundance for all da sites
Abundance_dyn <- c(butterfly_dynamics$Sabah,
butterfly_dynamics$Sarawak,
butterfly_dynamics$West.Kalimantan,
butterfly_dynamics$East.Kalimantan)
# wack together PGR for all sites
PGR <- c(butterfly_dynamics$PGR.Sabah,
butterfly_dynamics$PGR.Sarawak,
```

```
butterfly_dynamics$PGR.W.Kalimantan,
butterfly_dynamics$PGR.E.Kalimantan)
#big man gotta match the site factors
Site_dyn <- c(rep("Sabah",length(butterfly_dynamics$Sabah)),
rep("Sarawak",length(butterfly_dynamics$Sarawak)),
rep("West.Kalimantan",length(butterfly_dynamics$West.Kalimantan)),
rep("East.Kalimantan",length(butterfly_dynamics$East.Kalimantan)))
Site_dyn <- factor(Site_dyn)
#plot to see density dependance
plot(Abundance_dyn, PGR,
xlab = "Abundance (count per hour)",
ylab = "Population growth rate (PGR)")
#run the linear model to see if the PGR–abundance relationship differ between sites
m_resilience <- lm(PGR ~ Abundance_dyn * Site_dyn)
summary(m_resilience)
#now working out the carrying capacity per site
#big ups Statistics bio7 on the yt for the help doing: separate regressions per site
#sabah
m_Sabah <- lm(PGR.Sabah ~ Sabah, data = butterfly_dynamics)
coef(m_Sabah)
K_Sabah <- -coef(m_Sabah)[1] / coef(m_Sabah)[2]
K_Sabah
#sarawak
m_Sarawak <- lm(PGR.Sarawak ~ Sarawak, data = butterfly_dynamics)
coef(m_Sarawak)
K_Sarawak <- -coef(m_Sarawak)[1] / coef(m_Sarawak)[2]
K_Sarawak
#west kalimantan
m_WK <- lm(PGR.W.Kalimantan ~ West.Kalimantan, data = butterfly_dynamics)
coef(m_WK)
K_WestKalimantan <- -coef(m_WK)[1] / coef(m_WK)[2]
K_WestKalimantan
#moving onto the Easide (city where the sun dont set) (east kalimantan)
```

```
m_EK <- lm(PGR.E.Kalimantan ~ East.Kalimantan, data = butterfly_dynamics)

coef(m_EK)

K_EastKalimantan <- -coef(m_EK)[1] / coef(m_EK)[2]

K_EastKalimantan
```